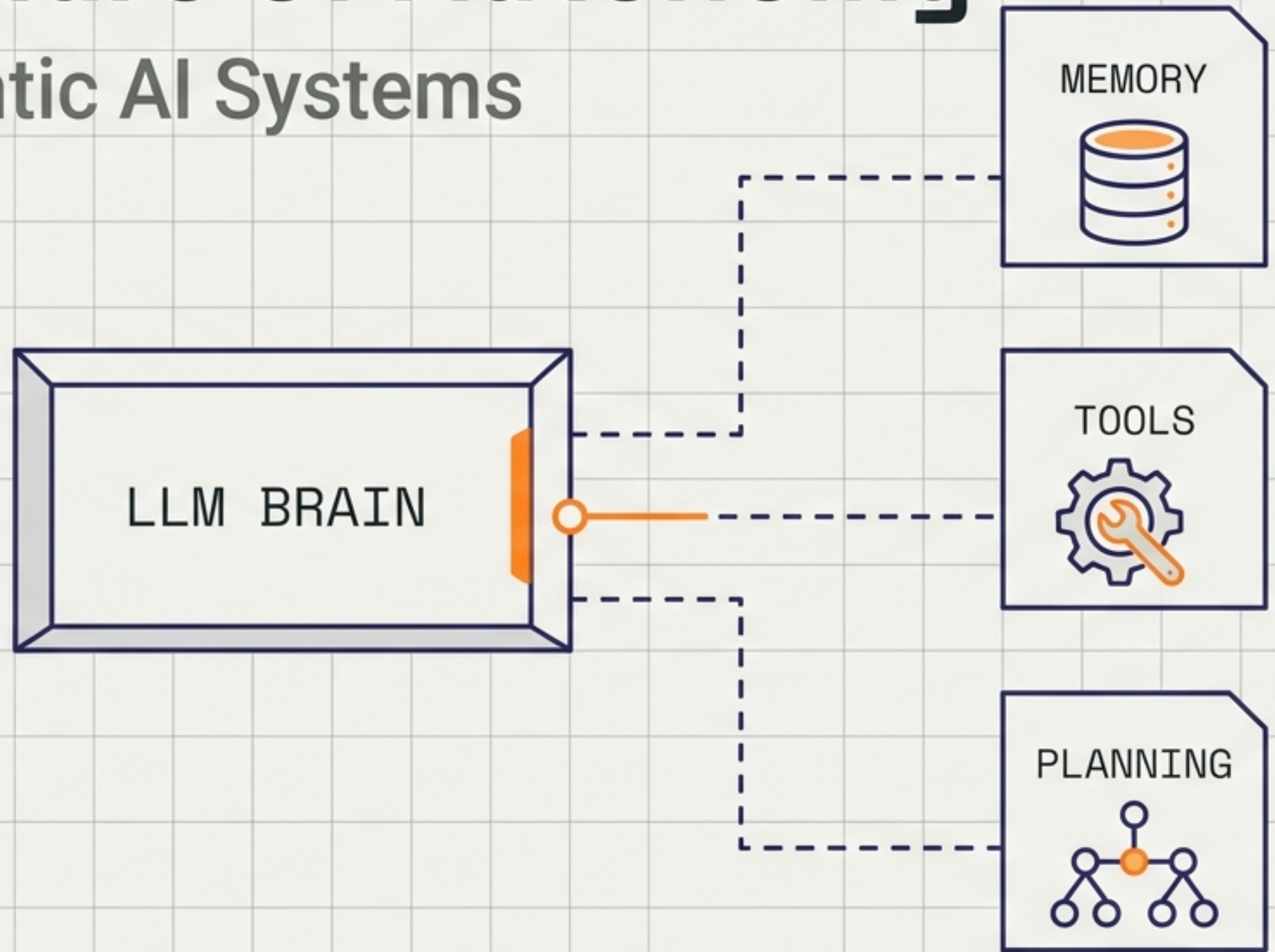
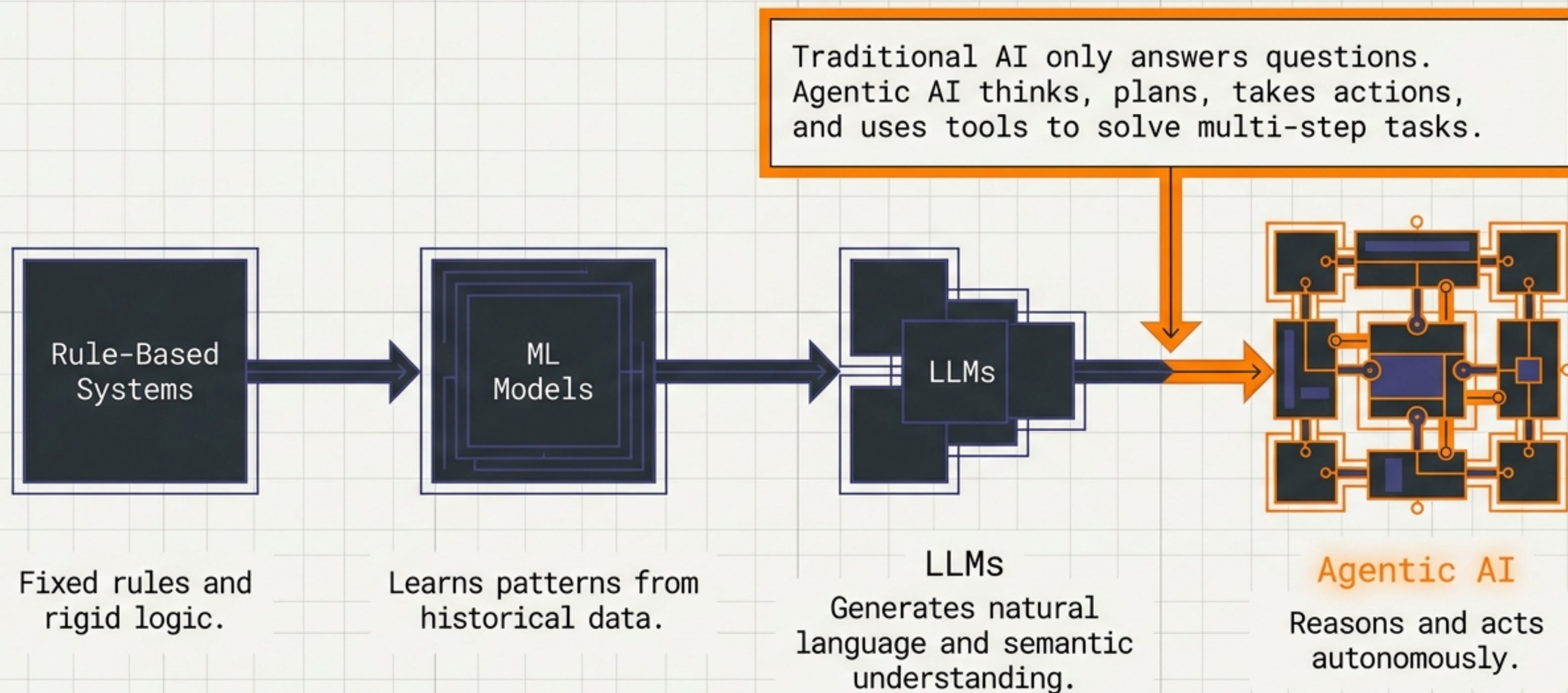


The Architecture of Autonomy

A Blueprint for Agentic AI Systems



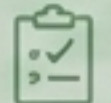
The Evolution from Generation to Execution



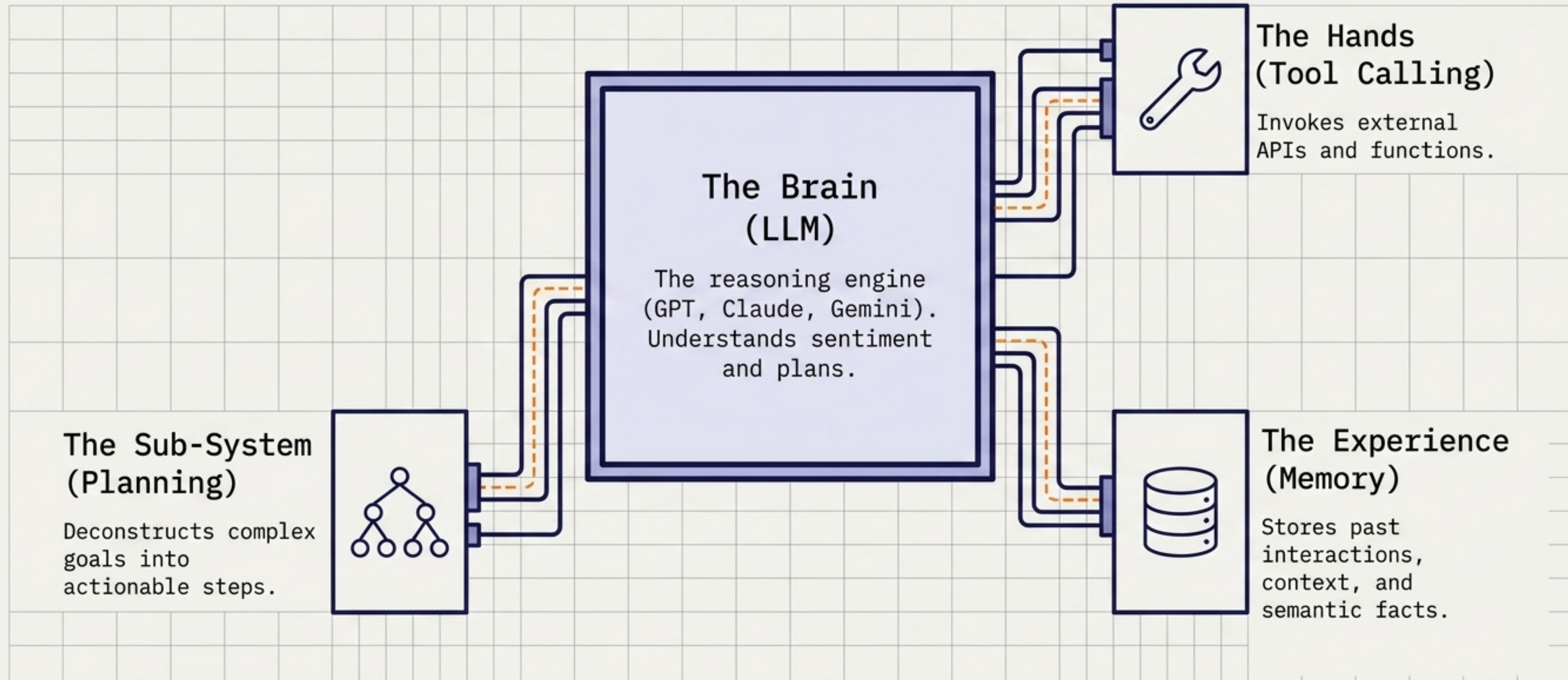
Passive Oracles vs. Autonomous Engines

Traditional Chatbot

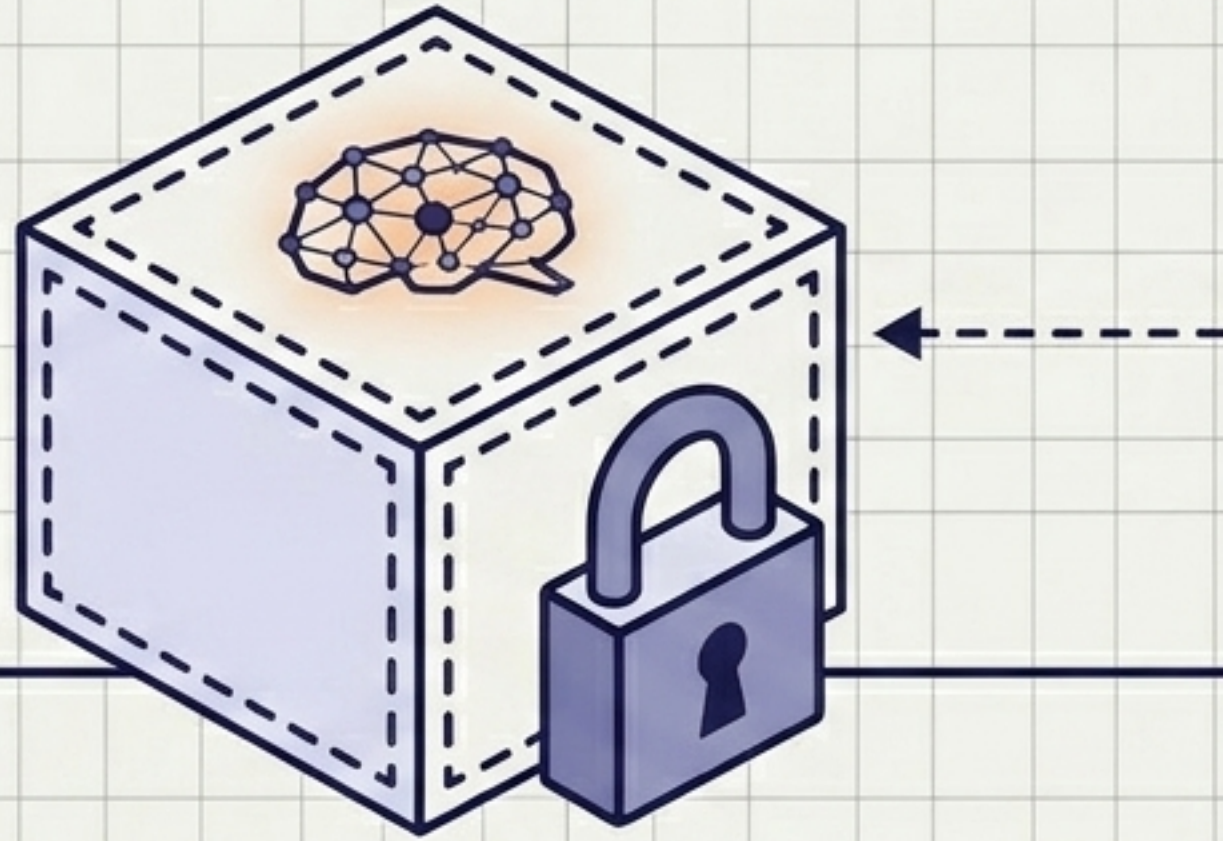
Agentic AI

System Behaviour	Passive, responds only when prompted.	Goal-oriented, works independently towards an objective.
Task Capability	Single-step text generation.	Breaks multi-step tasks into subtasks.
Environment Limits	Trapped in a closed box; zero external access.	Interacts with the real world via APIs and environments.
Example Request: "Book a flight"	Visit the airline website yourself.	Searches flights, compares prices, books tickets, and sends confirmation. 

The Anatomy of an Agentic Node



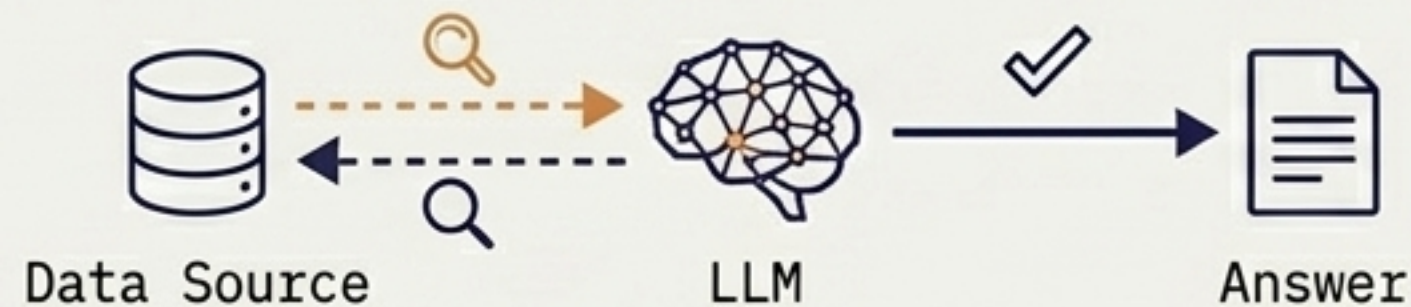
Escaping the Closed Box



Base models have zero real-world data and cannot perform tasks. They cannot read your database or browse the real-time internet.

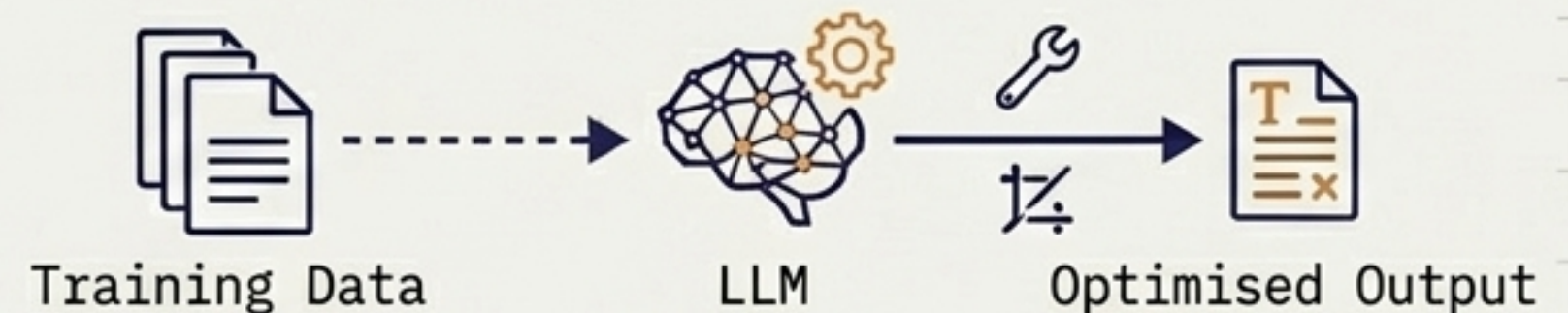
Retrieval-Augmented Generation (RAG)

Retrieves external info dynamically. Prevents hallucinations. Used for Memory.



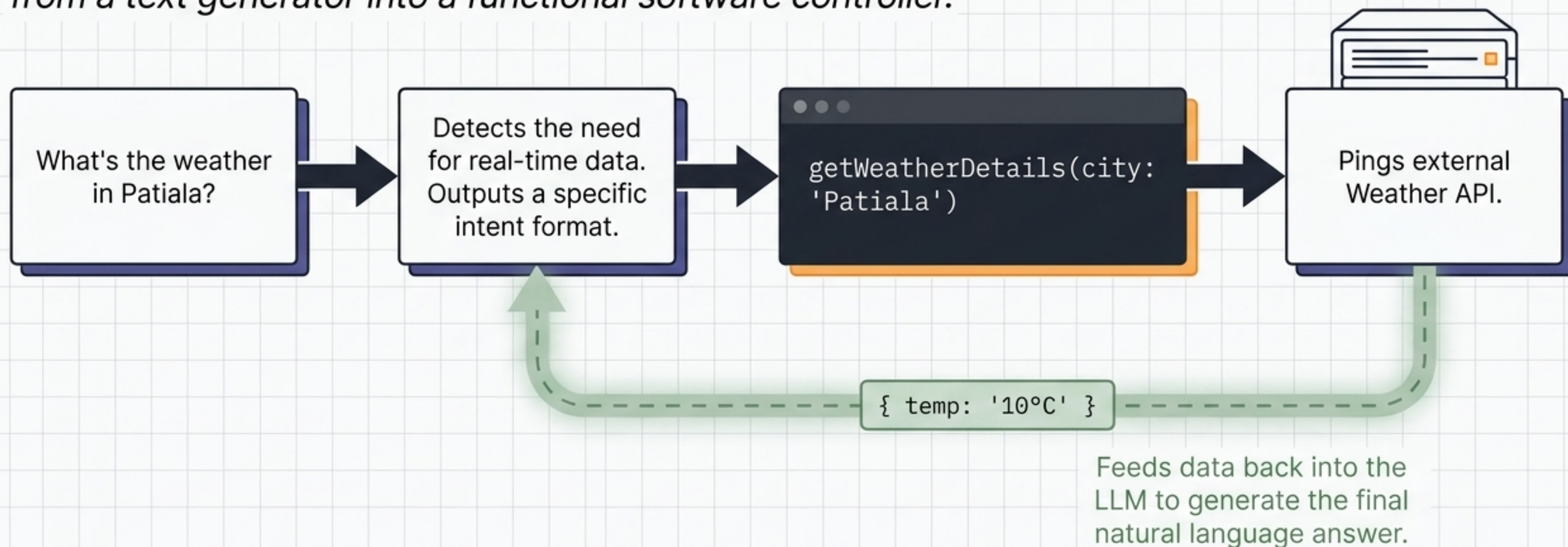
Fine-Tuning

Changes static model weights. Used for Specialised Formatting.

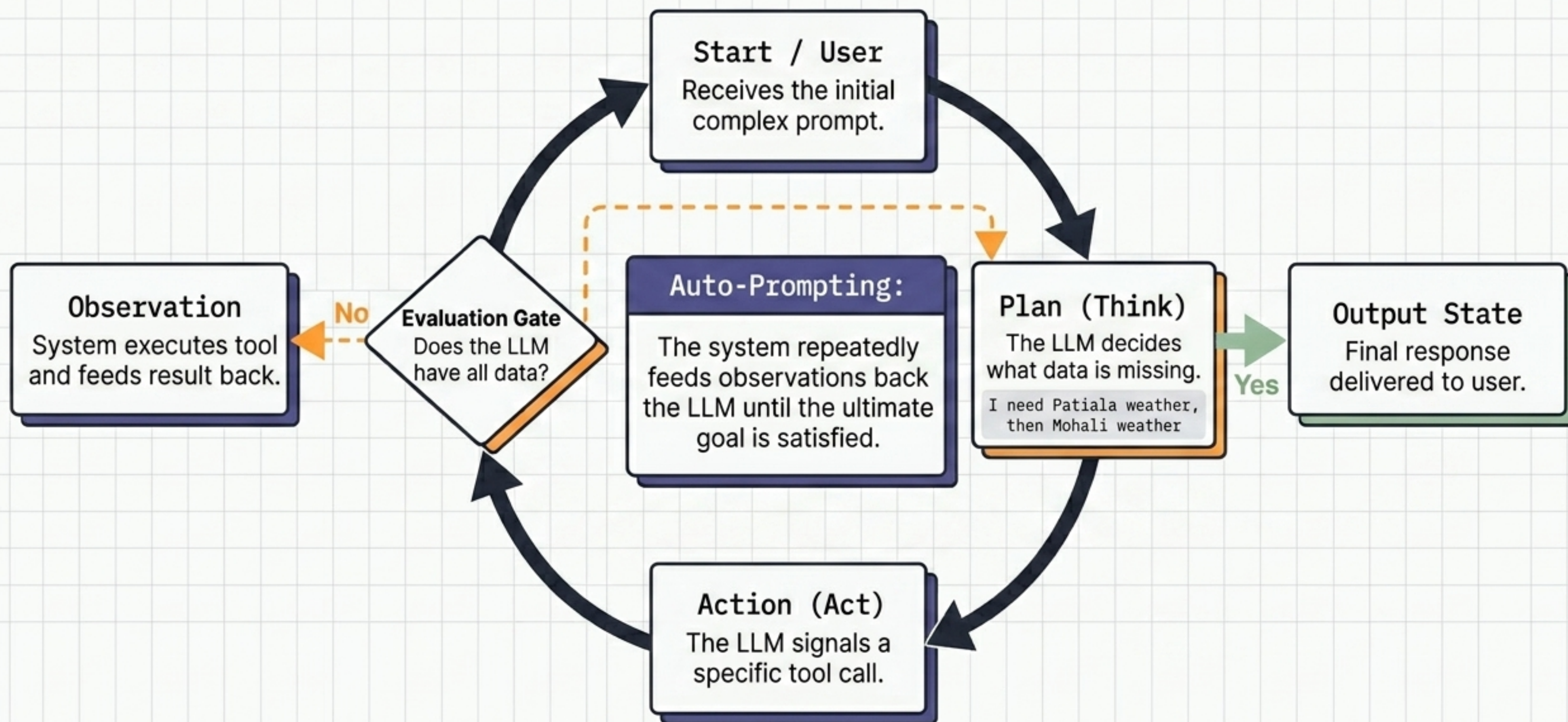


The Mechanics of Tool Calling

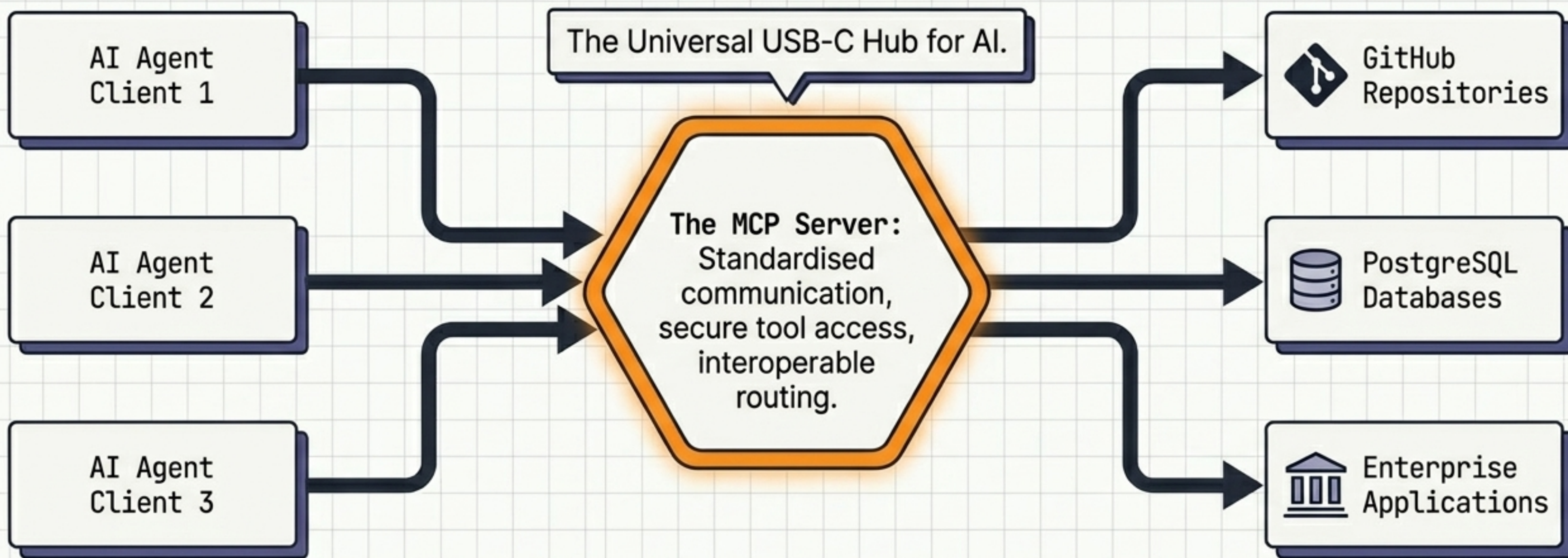
Tool calling is the engine of autonomy. It transforms a language model from a text generator into a functional software controller.



The ReAct State Machine (Reason + Act)

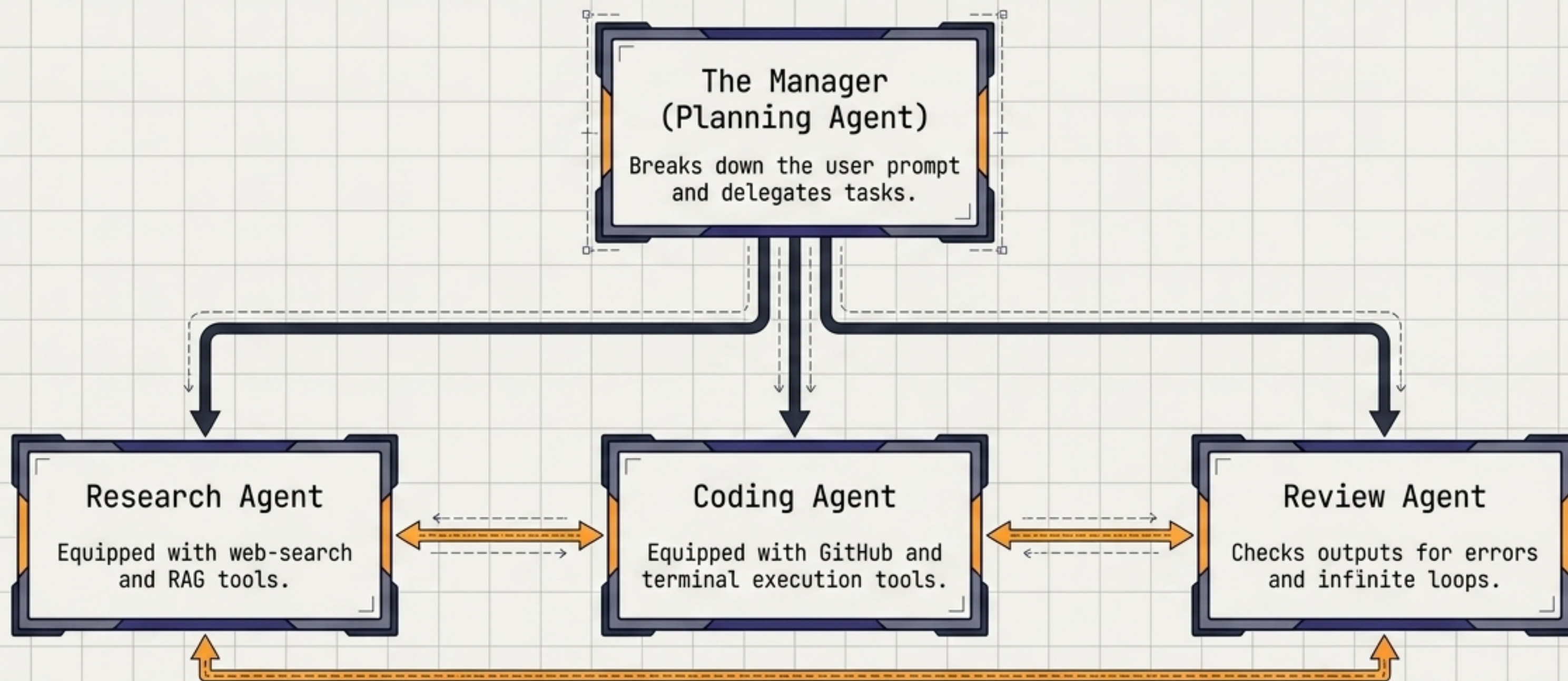


The Model Context Protocol (MCP)



Before MCP, every tool required custom integration. MCP provides a unified, secure interface for models to discover and invoke tools.

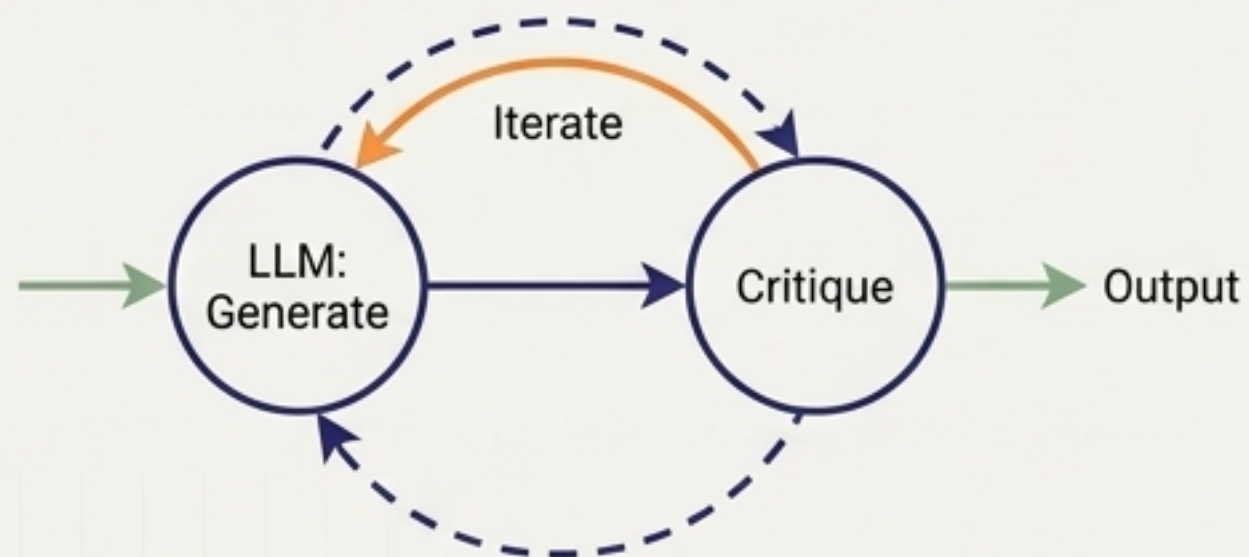
Agent-to-Agent (A2A) Orchestration



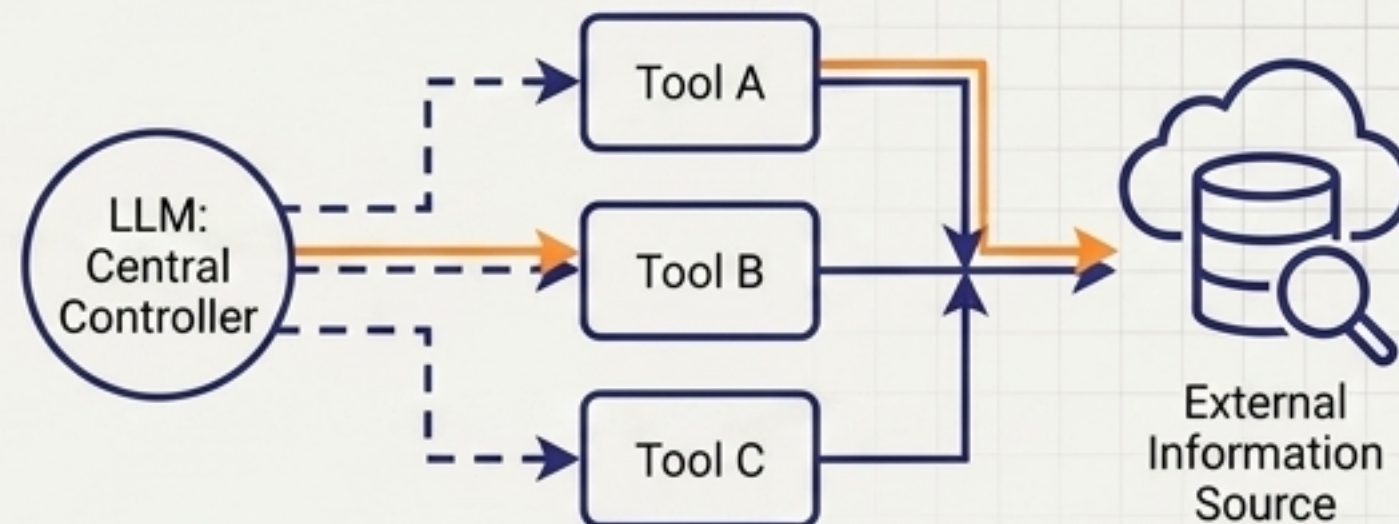
Frameworks like CrewAI and AutoGen bypass the limitations of single-agent overload by simulating parallel execution with specialised expertise.

Synthesis: Core Agentic Design Patterns

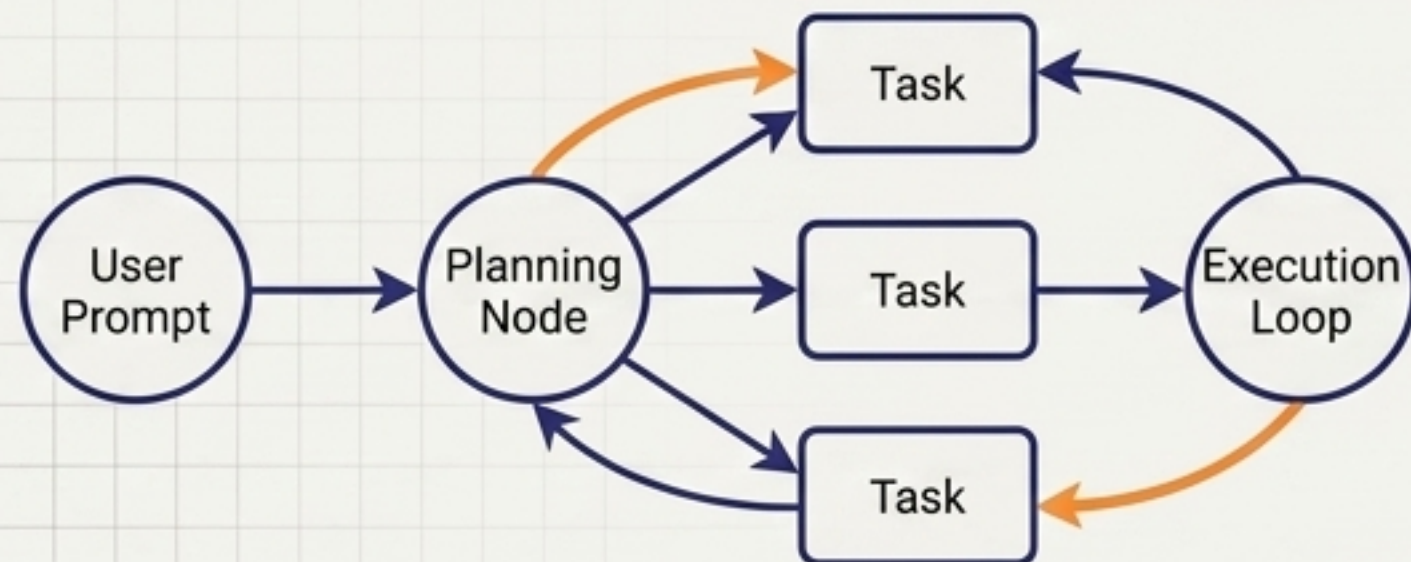
Reflection Pattern



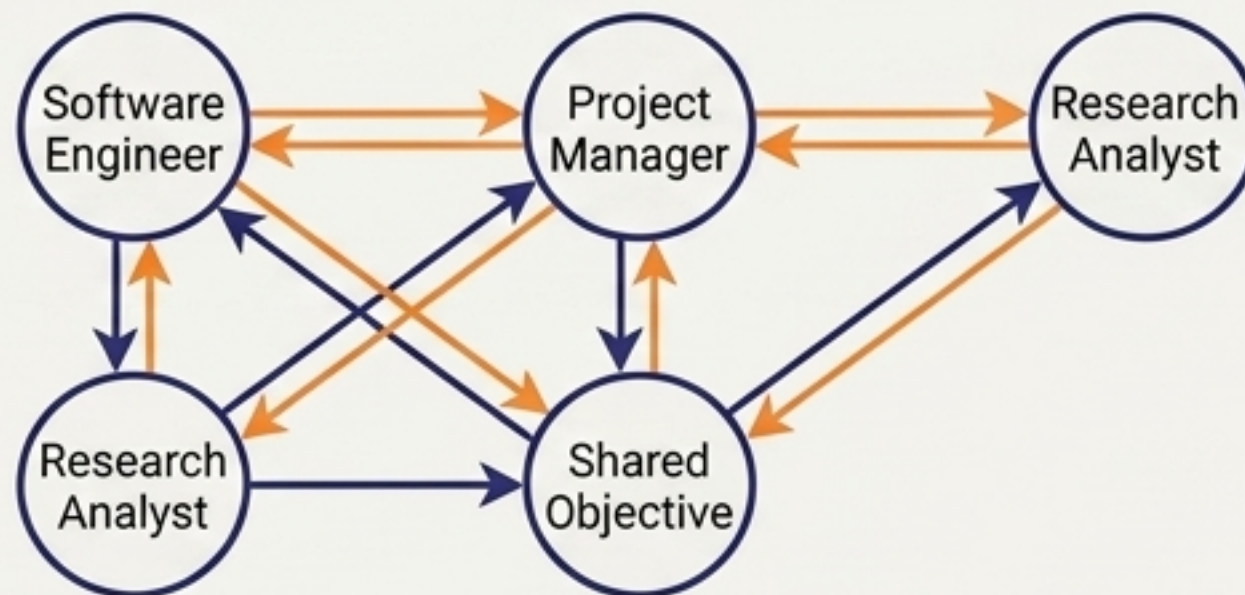
Tool Use Pattern



Planning Pattern



MultiAgent Pattern

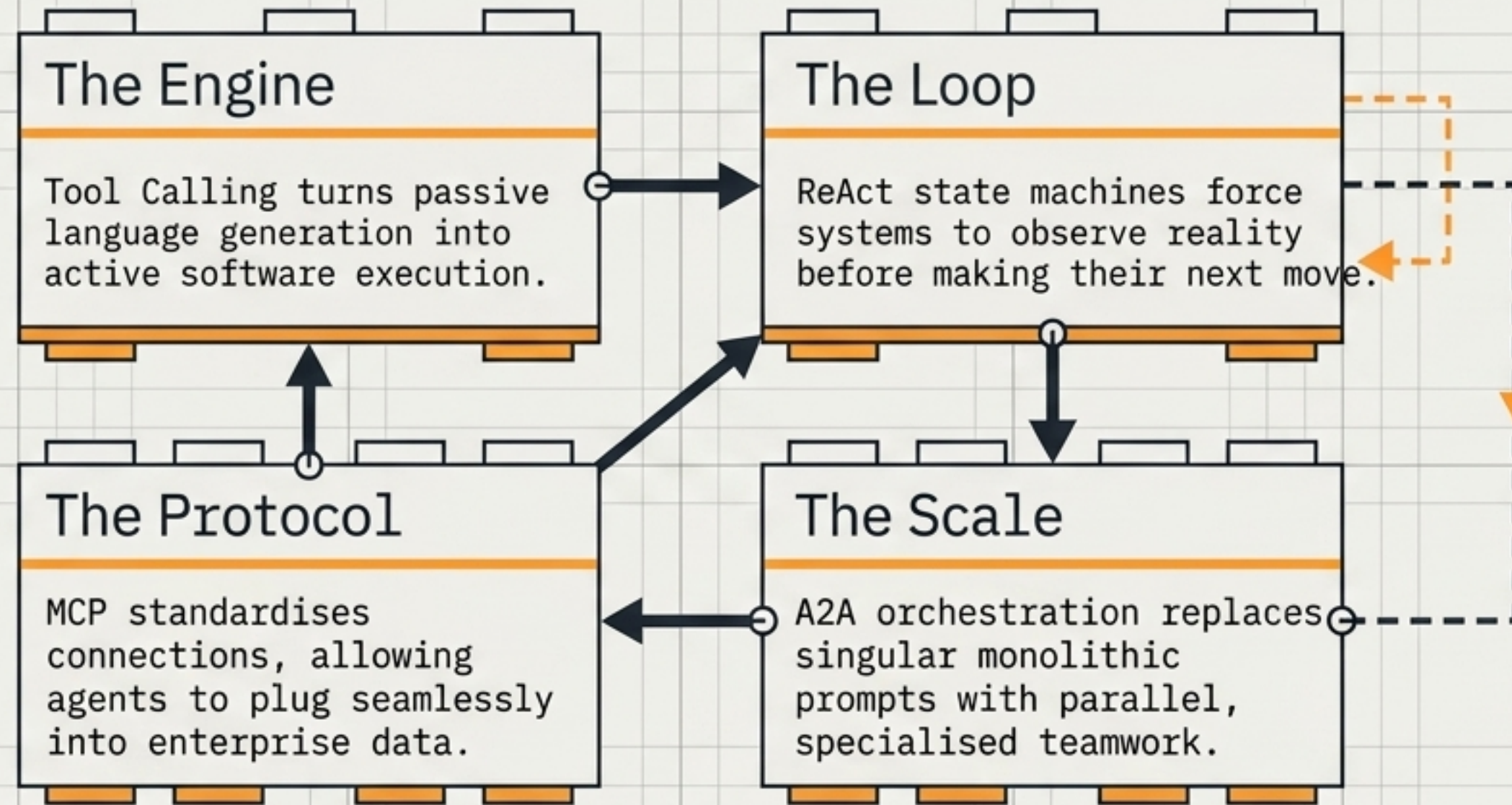


Friction Points and Security Boundaries



High-level autonomy demands strict approval boundaries. Critical APIs must never execute without explicit human authorisation.

The Intelligent Ecosystem



Agentic AI shifts the paradigm from software that advises to systems that execute.